MILP Formulation for Agroforestry Biomass Supply Chain (v10)

Multi-Cycle Harvesting with Age Lag Constraints

SmartAgroforst

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Contents

1 1. Index Sets

$$i \in \mathcal{I} = \{1, \ldots, n_s\} \qquad \text{Agroforestry sites} \tag{1}$$

$$j \in \mathcal{J} = \{1, \ldots, n_j\} \qquad \text{Storage/processing facilities} \tag{2}$$

$$k \in \mathcal{K} = \{1, \ldots, n_k\} \qquad \text{Consumer sites} \tag{3}$$

$$t \in \mathcal{T} = \{1, \ldots, T_{\max}\} \qquad \text{Time periods (years)} \tag{4}$$

$$t \in \mathcal{T}^{+} = \{0, \ldots, T_{\max} + 1\} \qquad \text{Time periods (years)} \tag{5}$$

$$t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} = \{A^{min} + 1, \ldots, T_{\max}\} \qquad \text{Harvesting periods (years)} \tag{6}$$

$$p \in \mathcal{P} = \{1, 2, 3\} \qquad \text{Product types (chemical, pulp, ener} \tag{7}$$

$$(s,t) \in \mathcal{S} = \left\{(s,t) | s,t \in 0, ..., (T_{\max} + 1) \wedge (t - s) \in \left\{A^{min}, ..., A^{max}\right\}\right\} \qquad \text{Consecutive harvesting} \tag{8}$$

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2 Example set $\mathcal{S}$

let $A^{min} = 1$, $A^{max} = 5$ and $T_{\max} = 10$, then $\mathcal{S}$ includes pairs like:

Subset	quantifier	Description
$(0, t)$	$t = 1, ..., 9$	establishment arcs
$(t, 11)$	$t = 1, ..., 10$	closing arcs
$(1, t)$	$t = 2, ..., 6$	cultivation arcs of period 1 (harvest in t)
$(2, t)$	$t = 3, ..., 7$	cultivation arcs of period 2 (harvest in t)
$(3, t)$	$t = 4, ..., 8$	cultivation arcs of period 3 (harvest in t)
$(4, t)$	$t = 5, ..., 9$	cultivation arcs of period 4 (harvest in t)
$(5, t)$	$t = 6, ..., 10$	cultivation arcs of period 5 (harvest in t)
$(6, t)$	$t = 7, ..., 10$	cultivation arcs of period 6 (harvest in t)
$(7, t)$	$t = 8, ..., 10$	cultivation arcs of period 7 (harvest in t)
$(8, t)$	$t = 9, ..., 10$	cultivation arcs of period 8 (harvest in t)
$(9, 10)$	$-$	cultivation arc of period 9 (harvest in 10)

If establishment occurs in period 3 and harvests in periods 6 and 10, the following variables would be active:

- $z_{i,0,3} = 1$ (establishment),
- $z_{i,3,6} = 1$ (harvest in 6 with establishment in 3),
- $z_{i,6,10} = 1$ (harvest in 10 with predecessor in 6),
- $z_{i,10,11} = 1$ (last harvest in 10),and
- $z_{ist} = 0$ for all $(s,t) \in \mathcal{S}$

Constraints C1 and C2 read like this

- C1: $\sum_{t=1}^{10} z_{i,0,t} \leq 1$ (at most one establishment) $\rightarrow$ as $z_{i,0,3} = 1$ it follows that $z_{i,0,t} = 0$ for all $t \neq 3$.
- C2_1: $z_{i,0,1} = \sum_{u \in 2,\ldots,6,11} z_{i,1,u}$ (if there is establishment in 1, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,1} = 0$, $\sum_{u \in 2,\ldots,6,11} z_{i,1,u} = 0$.
- C2_2: $z_{i,0,2} + z_{i,1,2} = \sum_{u \in 3,\ldots,7,11} z_{i,2,u}$ (if there is establishment or harvest in 2, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,2} = z_{i,1,2} = 0$, $\sum_{u \in 3,\ldots,7,11} z_{i,3,u} = 0$
- C2_3: $z_{i,0,3} + z_{i,1,3} + z_{i,2,3} = \sum_{u \in 4,\ldots,8,11} z_{i,3,u}$ (if there is establishment or harvest in 3, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,3} = 1$ and $z_{i,1,3} = z_{i,2,3} = 0$, $\sum_{u \in 4,\ldots,8,11} z_{i,3,u} = 1 \rightarrow$ as $z_{i,3,6} = 1$, it follows that $z_{i,3,u} = 0$ for all $u \neq 6$.
- C2_4: $z_{i,0,4} + z_{i,1,4} + z_{i,2,4} + z_{i,3,4} = \sum_{u \in 5,\ldots,9,11} z_{i,4,u}$ (if there is establishment or harvest in 4, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,4} + z_{i,1,4} + z_{i,2,4} + z_{i,3,4} = 0$, it follows that $\sum_{u \in 5,\ldots,9,11} z_{i,4,u} = 0$.
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3 2. Decision Variables

3.1 2.1 Binary Variables

$$z_{ist} \in \{0,1\} \quad \text{Indicator if at site } i \text{ in period} t, s \text{ are consecutive harvests with } (s,t) \in \mathcal{S} \tag{9}$$

3.2 2.2 Continuous Variables

$$X_{ijpt} \geq 0 \quad \text{Flow of product } p \in \mathcal{P} \text{ from site } i \in \mathcal{I} \rightarrow \text{ storage } j \in \mathcal{J} \text{ in period } t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \text{ [tonnes]} \tag{10}$$

$$X_{jkpp't} \geq 0 \quad \text{Conversion flow of product } p \in \mathcal{P} \text{ to product } p' \in \mathcal{P} : p' \geq p \text{ from storage } j \rightarrow \text{ consumer } k \text{ in peri} \tag{11}$$

$$Y_{ipt} \geq 0 \quad \text{Maximum harvest quantity of product } p \text{ at site } i \text{ in period } t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \text{ [tonnes]} \tag{12}$$

$$S_{jpt} \geq 0 \quad \text{Inventory at storage } j \text{ in period } t \in \mathcal{T} \text{ [tonnes]} \tag{13}$$

4 3. Parameters

Parameter	Unit	Description	Typical Value
A^{min}	yr	minimum age	3–5 years
A^{max}	yr	maximum age	8–10 years
AREA_i	ha	Site area	8–20 ha
c_i^{est}	€/ha	Establishment cost in € per ha	10 €/ha
c_i^{opp}	€/ha	Opportunity cost per ha	5 – 15 €/ha
c_i^{harv}	€/ha	Harvest and refitting cost per ha	5 – 15 €/ha
η_{pa}	t/ha	Base yield for product p, age a	4–60 t/ha
$R_{k,p}$	€/t	Revenue for product p by consumer k	80–250 €
$c^{tr-raw/pre}$	€/t/km	Transport cost rates per ton and kilometer for raw and pre-treated biomass	0.10/0.15 €/km-ton
d_{ij}, d_{jk}	km	Distances	30–200 km
CAP_j^{stor}	t	Storage capacity	300–500 t
CAP_j^{proc}	t/yr	Processing capacity	80–150 t/yr
D_{kp}^{max}	t/yr	Demand at consumer k	Varies

5 4. Objective Function

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\begin{aligned}
\max Z = \quad & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} && \text{Revenue} && (14)\\
& - \sum_{i} C_i^{est} \cdot \text{AREA}_i \cdot \sum_{t \in T} z_{i0t} && \text{Establishment} && (15)\\
& - \sum_{i} C_i^{opp} \cdot \text{AREA}_i \cdot \sum_{t \in T} (T^{max} - t) \cdot z_{i0t} && \text{Opportunity} && (16)\\
& - \sum_{i,(s,t) \in \mathcal{S} | s > 0} C_i^{harv} \cdot \text{AREA}_i \cdot z_{ist} && \text{Harvest} && (17)\\
& - c^{tr-raw} \cdot \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} && \text{Transport (site}\rightarrow\text{hub)} && (18)\\
& - c^{tr-pre} \cdot \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} && \text{Transport (hub}\rightarrow\text{consumer)} && (19)\\
& - \sum_{j,p,t} c_j^{stor} \cdot S_{jpt} && \text{Storage} && (20)
\end{aligned}
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6 5. Constraints

6.1 Constraint Set 1: AFS establishment

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \qquad \forall i \in \mathcal{I} \tag{C1}$$

Interpretation: If AFS is established in t, $z_{i,0,t} = 1$.

6.2 Constraint Set 2: Path connectivity

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \qquad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{C2}$$

Interpretation: If there is a harvest in t, there must be an predecessor in $s < t$ and successor in $u > t$.

6.3 Constraint Set 3: Biomass yield calculation

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot AREA_i \cdot z_{ist} \qquad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C3}$$

(21)

Interpretation: Biomass yield is restricted by age of plantage

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6.4 Constraint Set 4: Yield Shipping Linkage

$$\sum_{j \in \mathcal{J}} X_{ijpt} \le Y_{ipt} \qquad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C4}$$

Interpretation: Biomass shipped from site i (product p, period t) is limited by available harvest.

6.5 Constraint Set 5: Inventory Balance

$$S_{jpt} = S_{jp(t-1)} + \sum_{i} X_{ijpt} - \sum_{k,p' \ge p} X_{jkpp't} \qquad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C5}$$

with $S_{j,p,0} = 0$.

Interpretation: Flow conservation at storage hubs.

6.6 Constraint Set 6: Storage Capacity

$$\sum_{p \in \mathcal{P}} S_{jpt} \le \mathrm{CAP}_j^{stor} \qquad \forall j \in \mathcal{J}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C6}$$

Interpretation: Total inventory limited by physical storage space.

6.7 Constraint Set 7: Processing Capacity

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \le \mathrm{CAP}_j^{proc} \qquad \forall j \in \mathcal{J}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C7}$$

Interpretation: Incoming material throughput limited by processing equipment.

6.8 Constraint Set 8: Demand Satisfaction with Product Cascade

$$\sum_{j \in \mathcal{J}, p' \in \mathcal{P} : p' \leq p} X_{jkp'pt} \leq D^{max}_{kpt} \qquad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{\langle\dashv\nabla\sqsubseteq} \tag{C8}$$

Interpretation: Higher-quality products (lower p) can substitute for lower-quality demand.

Cascade hierarchy: - Product 1 (chemical) $\rightarrow$ Satisfies any demand - Product 2 (pulp) $\rightarrow$ Satisfies product 2 or 3 demand - Product 3 (energy) $\rightarrow$ Only satisfies product 3 demand

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